EXIT TICKET / REFLECTION

## Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Every virtual machine must run a complete guest operating system.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

## Lesson 111 · OS Types, Boot Firmware, Drivers and VMs

Name: _____ Class: _____ Date: _____

**1. In round robin, what happens when a running process's quantum expires but it is unfinished?**

**2. After pushing A then B onto a stack, which is popped first?**

**3. Why must a binary search operate on ordered data?**

**4. An OR gate has inputs 0 and 1. What is the output?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

It returns to the ready queue (subject to scheduler rules).

**2. Model answer**

B, because a stack is LIFO.

**3. Model answer**

It discards half the search space using an ordering comparison.

**4. Model answer**

1.

**By the end, you can…**

• Embedded/real-time emphasise constraints; multi-user and distributed coordinate shared work.

• BIOS/UEFI firmware begins startup; drivers provide hardware-specific access.

• A hypervisor runs guest OSs; a language VM executes intermediate code.

**Vocabulary — know these terms**

real-time OS — System where meeting time constraints is essential.

BIOS/UEFI — Platform firmware that begins boot preparation.

hypervisor — Manages virtual machines running guest OSs.

device driver — Hardware-specific interface supplied to an OS.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

Compare a hypervisor virtual machine with a bytecode virtual machine and explain a real-time deadline. [8]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 1**

Explain two differences between multi-user and multitasking operating systems, giving examples. [6]

**Question 2**

Describe the roles of BIOS/UEFI firmware and a device driver during startup and use. [6]





NEW LEARNING

# Read, annotate and explain

**Match OS to use**

Embedded/real-time emphasise constraints; multi-user and distributed coordinate shared work.

**Boot and control**

BIOS/UEFI firmware begins startup; drivers provide hardware-specific access.

**Virtualise**

A hypervisor runs guest OSs; a language VM executes intermediate code.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**An industrial safety controller must respond within a guaranteed deadline. Choose an OS type and justify it.**

### Model answer

A hard real-time operating system is suited because missing a deadline is unacceptable; deterministic scheduling/latency matters more than high average throughput. An embedded implementation may also apply.

Check: link each claim to the scenario and use precise OCR terminology.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve